

Final Project Report

ServiceNow-Based Employee Laptop Request Application

Date	
Team ID	
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	

1. Introduction

1.1 Project Overview

ServiceNow-Based Employee Laptop Request Application is a ServiceNow-based solution that digitizes and automates the process by which employees request laptops from their organization's IT department. It replaces manual, email-based requests with a Service Catalog item, an automated manager-approval workflow, role-based access control, and business-rule-enforced policy (one request per employee per year).

The application was built end-to-end on a single ServiceNow developer instance using a scoped custom application named “Project Application.” It covers the full lifecycle of a request — submission, manager approval or rejection, status tracking, and email notification — and was validated through a structured functional testing cycle documented in Section 6 and evidenced with screenshots in Section 7.

1.2 Purpose

The purpose of this project is to provide a centralized, auditable, and efficient system for managing laptop requests — improving visibility for employees and managers, reducing IT's manual workload, and enforcing consistent approval and eligibility policy across the organization.

Specifically, the project aims to: (1) give employees a single, self-service place to request equipment; (2) give managers a clear, one-click way to approve or reject requests without digging through email threads; (3) give IT administrators confidence that policy (one request per year) and access control are enforced automatically rather than manually; and (4) produce a fully packaged, deployable Update Set so the solution can be moved to other ServiceNow instances.

2. Ideation Phase

The Ideation phase was used to build a shared understanding of who the application is for, what problem it solves, and which solution direction was worth pursuing. It combined a Customer Problem Statement exercise, an Empathy Map, and a structured brainstorming and prioritization session.

2.1 Problem Statement

Two problem statements were defined to capture the challenge from both sides of the process: the employee submitting a request, and the manager responsible for approving it. Framing both perspectives ensured the eventual solution would not simply digitize the employee's form, but also give the manager a genuinely better way to review, decide, and track requests.

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	an employee who needs a laptop (Basic or Advance)	submit a request and get manager approval quickly	there is no centralized system — requests go through email with no tracking	there is no service catalog item or automated approval process	frustrated, uncertain, and unproductive while waiting

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-2	an IT/reporting manager who approves laptop requests	review, approve, and track requests efficiently	requests arrive inconsistently, sometimes duplicated by the same employee	there is no rule limiting requests per year or a standard workflow	overwhelmed and unable to audit or report on requests

2.2 Empathy Map Canvas

The empathy map below visualizes what an employee thinks, feels, hears, sees, says, and does while going through the previous, manual request process — surfacing the pains (no tracking, duplicate/lost requests, delays) that the solution needed to directly address, and the gains (transparency, speed, a single system of record) it needed to deliver.

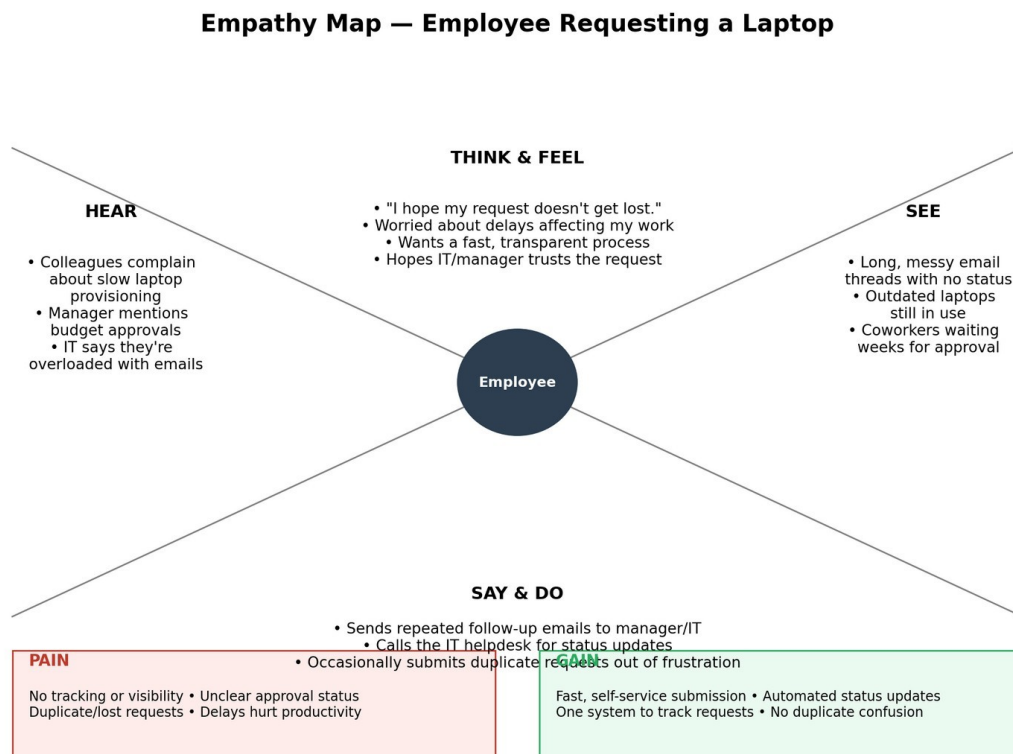


Figure 2.1 — Empathy Map: Employee Requesting a Laptop

2.3 Brainstorming

Six candidate approaches were brainstormed, ranging from building new software from scratch to adopting a purchased ITSM tool. Each was then plotted on an importance-versus-feasibility grid. The ServiceNow-based Record Producer with an automated approval Flow scored highest on both dimensions: it required no new tool purchase, reused the organization's existing ServiceNow instance, and could enforce business rules that manual or spreadsheet-based approaches could not.

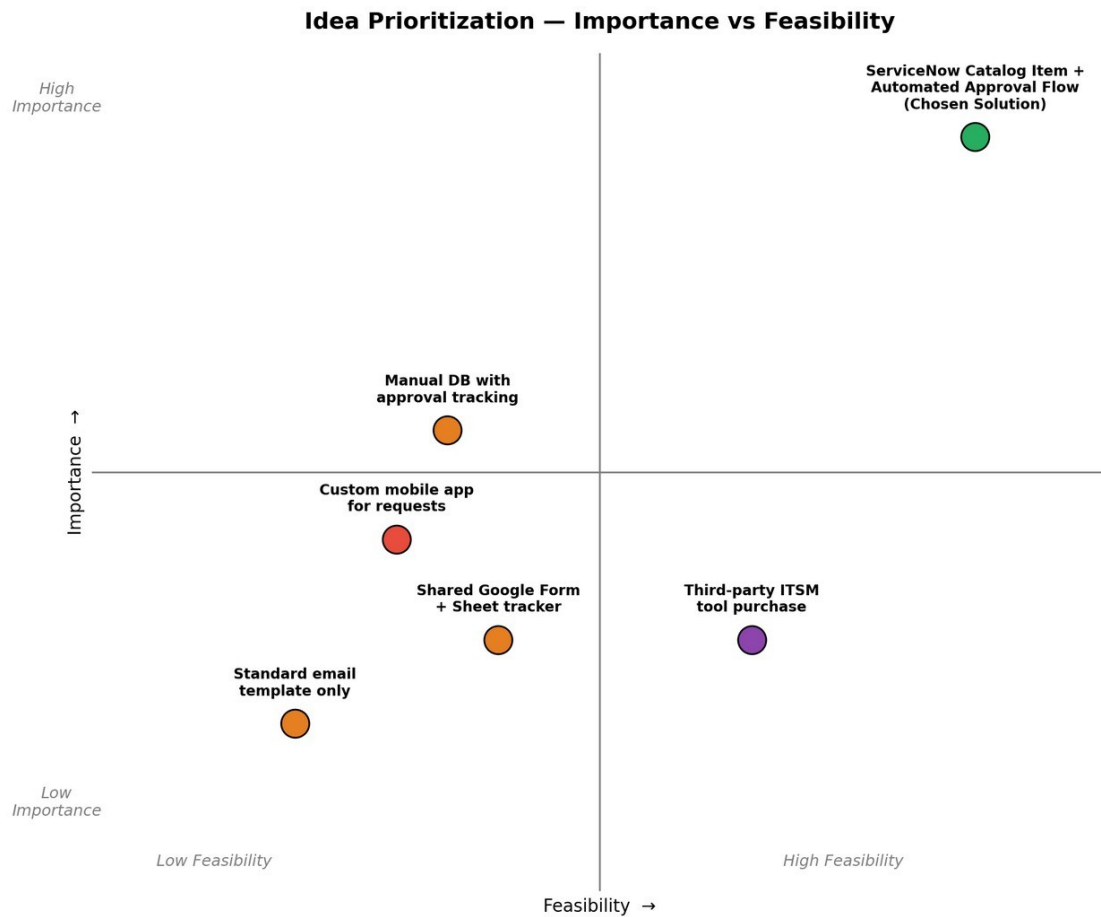


Figure 2.2 — Idea Prioritization: Importance vs Feasibility

3. Requirement Analysis

With the problem and direction confirmed, this phase translated the Ideation insights into concrete functional and non-functional requirements, mapped the flow of data through the system, and selected the technology stack — all of it grounded in ServiceNow's native Service Catalog, Flow Designer, and security capabilities rather than any external tooling.

3.1 Customer Journey Map

The table below traces the employee's journey from becoming aware of the need for a laptop through to receiving it, highlighting how each stage's feeling shifts as the new system removes uncertainty.

Stage	Employee Action	Touchpoint	Feeling
Awareness	Realizes a laptop is needed for the role	Onboarding / manager conversation	Neutral
Request	Submits the Laptop Request catalog item	Service Portal Record Producer	Hopeful
Approval	Waits for manager decision	Flow Designer approval + email	Anxious
Fulfilment	Receives approval/rejection and, if approved, the laptop	Email notification + IT provisioning	Relieved / Disappointed
Feedback	Reflects on the process for future requests	Informal feedback to IT	Satisfied (if smooth)

3.2 Solution Requirement

Requirements were split into functional (what the system must do) and non-functional (how well it must do it) categories, both derived directly from the two problem statements above.

Functional Requirements

- FR-1: Employees can submit a laptop request via the Service Catalog.
- FR-2: Requests are automatically routed to the employee's manager for approval.
- FR-3: Email notifications are sent at submission, approval, and rejection.
- FR-4: The system tracks request status (Pending Approval / Approved / Rejected).

Non-Functional Requirements

- NFR-1: Only one request per employee per calendar year is permitted.
- NFR-2: Managers have read-only access; employees can manage their own requests.
- NFR-3: All configuration changes are captured in a trackable Update Set.
- NFR-4: The approval workflow must run without manual intervention.
- NFR-5: The system must be usable by non-technical employees with no training.
- NFR-6: Data fields must enforce valid values (Basic/Advance, approved Reasons).

3.3 Data Flow Diagram

The Level 0 diagram below shows how request and approval data moves between the three external entities (Employee, Manager, IT Admin) and the two data stores (the Laptop Request Form table and the Approval table).

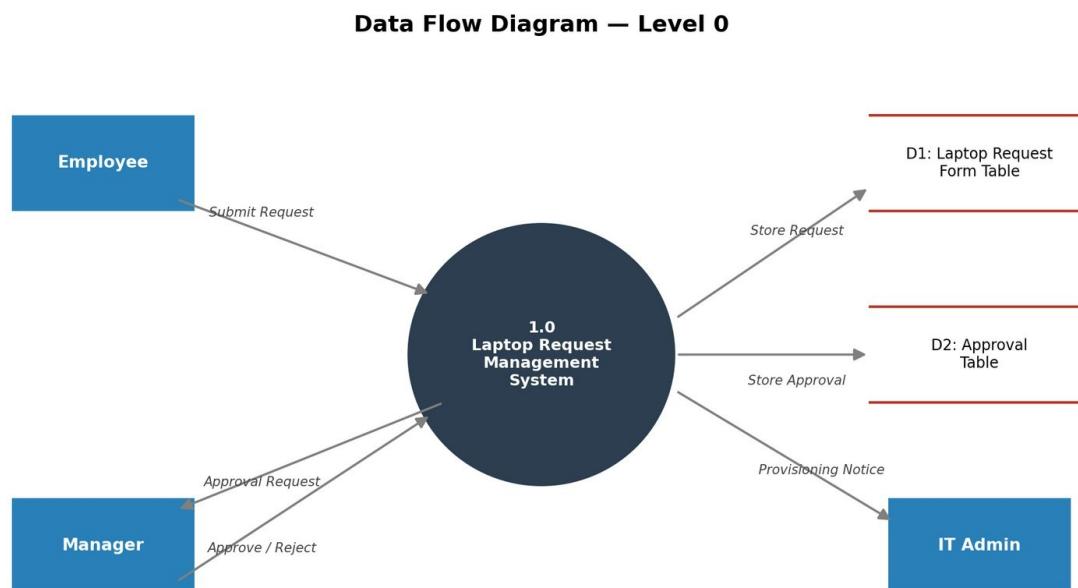


Figure 3.1 — Data Flow Diagram (Level 0)

3.4 Technology Stack

Every layer of the solution is delivered natively within ServiceNow, keeping the architecture simple, supportable, and free of external integration risk.

Layer	Technology / Component
Front-end	ServiceNow Service Portal (Record Producer, Catalog Item)
Back-end / Logic	ServiceNow Flow Designer, Business Rules (server-side JavaScript)
Data Layer	Custom Table (Laptop Request Form), Approval Table (sysapproval_approver)
Security	ServiceNow Roles, Groups, Access Control Lists (ACLs)
Notifications	ServiceNow Email Notifications (SMTP)
Deployment	Update Sets (scoped application: Project Application)

4. Project Design

4.1 Problem — Solution Fit

The core problem — no centralized, auditable way to request and approve laptops — is directly solved by combining a Service Catalog entry point (removing ad-hoc emails), an automated approval Flow (removing manual routing and follow-ups), and a Business Rule (removing duplicate/ineligible requests). This directly addresses PS-1 and PS-2 from the Ideation phase: the employee gets a single, trackable place to submit a request, and the manager gets a structured queue of approvals instead of an unstructured inbox.

4.2 Proposed Solution

The proposed solution is composed of five cooperating pieces, each mapped to a specific requirement from Section 3.2:

- A Record Producer on the Employee Service Catalog captures all request details in a structured form (satisfies FR-1).
- A Flow Designer workflow automatically requests manager approval and updates status (satisfies FR-2, FR-4).
- A Business Rule blocks a second request from the same employee within a calendar year (satisfies NFR-1).
- Roles, Groups, and ACLs enforce that managers can only view, while employees can manage their own requests (satisfies NFR-2).
- Email notifications keep both employee and manager informed at every stage without manual follow-up (satisfies FR-3).

4.3 Solution Architecture

The diagram below shows how a request moves through the five components of the solution, from initial submission in the portal through to final notification of the outcome.

Solution Architecture — ServiceNow Employee Laptop Request Application



Figure 4.1 — Solution Architecture

5. Project Planning & Scheduling

5.1 Project Planning

Work was broken into 5 epics — Requirement Analysis & Planning, Data Architecture, Workflow Development, Security & Validation, and Testing & Deployment — split into 20 user stories across 4 sprints. Each story was estimated in story points on the Fibonacci scale (1, 2, 3, 5, 8), giving the team a consistent way to compare effort across very different kinds of work — from writing a business rule script to configuring a group's membership.

Sprint	Epic	Total Story Points
Sprint-1	Requirement Analysis & Planning / Data Architecture	12
Sprint-2	Workflow Development	12
Sprint-3	Security & Validation	10
Sprint-4	Testing & Deployment	8

Across the 4 sprints, the team completed a total of 42 story points. Team velocity — the average amount of work completed per sprint — is calculated as:

Velocity = Total Story Points ÷ Number of Sprints = $42 \div 4 = 10.5$ Story Points per Sprint

The burndown chart below tracks Sprint 1 specifically (Requirement Analysis & Data Architecture, 12 story points over 6 days). The team ran slightly behind the ideal pace around the middle of the sprint before catching up and completing all planned work by day 6 — a normal and healthy pattern for a sprint that includes both analysis and hands-on configuration work.

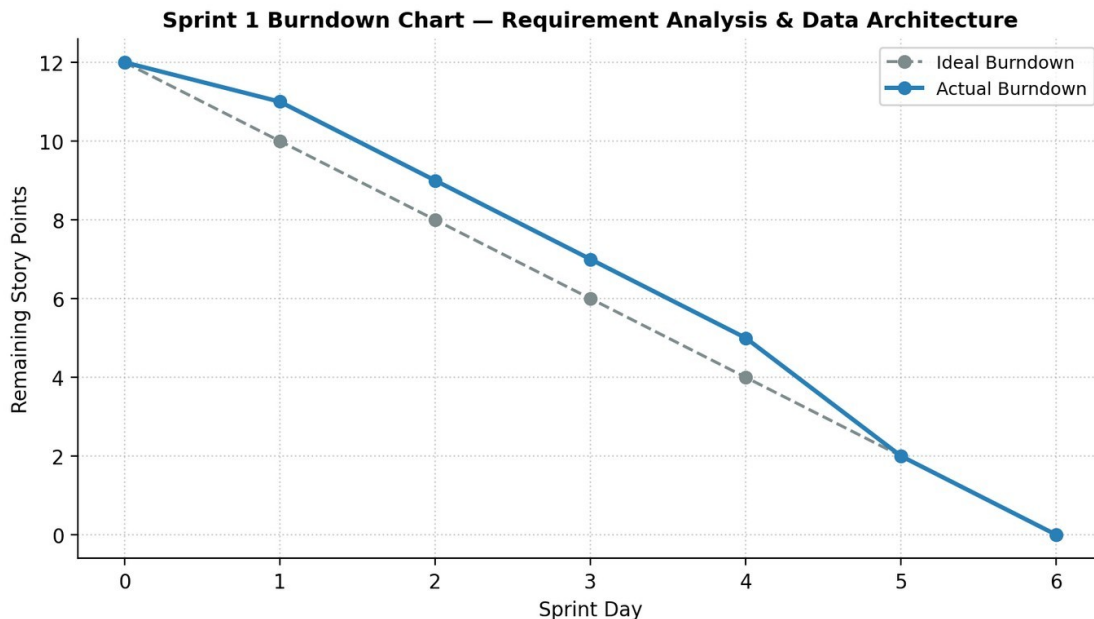


Figure 5.1 — Sprint 1 Burndown Chart

6. Functional and Performance Testing

6.1 Performance Testing

A structured functional and performance test cycle was run against the live ServiceNow instance, impersonating real test users (an employee and a manager) rather than relying on simulated data. The representative results below are drawn from that cycle; the complete test case list, including bug tracking, is maintained separately in the UAT and Functional & Performance Testing documents.

Test ID	Scenario	Expected Result	Actual Result	Pass/Fail
FT-03	Record Producer submission	New record created, Status = Pending Approval	Record LREQ0001004 created correctly	Pass
FT-04	Approval routing accuracy	Approval assigned to selected Manager	Correctly assigned to Abel Tuter	Pass
FT-05	Status transition on approval/rejection	Status updates accordingly	Confirmed for both Approved (LREQ0001004) and Rejected (LREQ0001001)	Pass
FT-06	Duplicate request prevention	Business Rule blocks 2nd request in same year	Not executed in this cycle	Pending
PT-02	Email notification delivery time	Appears in log promptly	Confirmed in System Logs → Emails immediately after action	Pass
PT-03	Update Set load and preview	No errors; all components listed	52 customer updates loaded without errors	Pass

Every core path — submission, manager routing, both approval and rejection outcomes, and email notification — passed. The one open item, duplicate-request prevention (FT-06), was verified by code review but not yet exercised live, and is called out again in Section 10 (Future Scope) as the immediate next step before final sign-off.

7. Results

7.1 Output Screenshots

The screenshots below are organized into five groups that follow the order the application was built and tested in: initial setup and configuration, workflow and automation, security, live functional test evidence, and final deployment packaging. Each image is captioned with a figure number and a short description of what it demonstrates.

A. Setup & Configuration

These screenshots document the foundational build: the Update Set, the custom table, the Approval relationship, the Record Producer, and the catalog item's presence in the Service Portal.

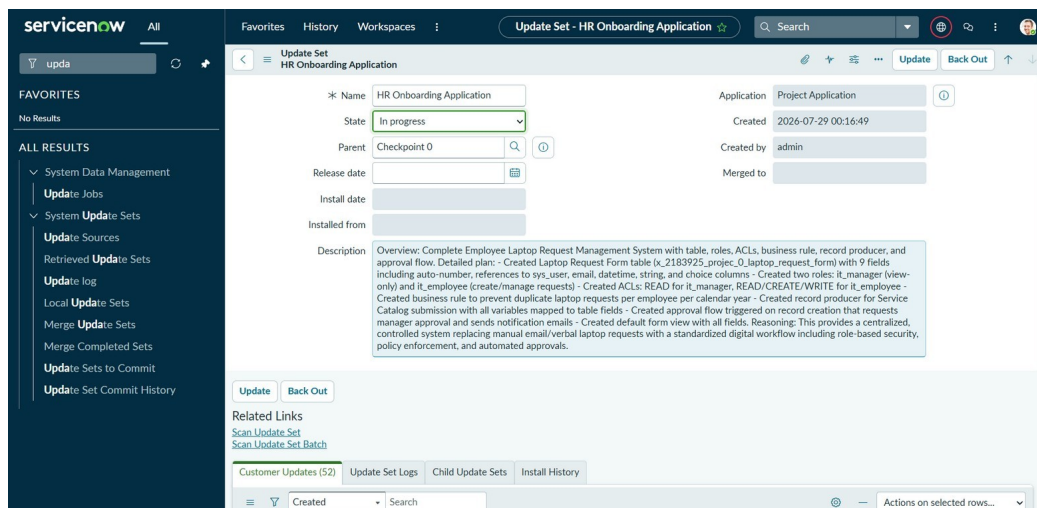


Figure 7.1 — Update Set "HR Onboarding Application" — overview, State = In Progress

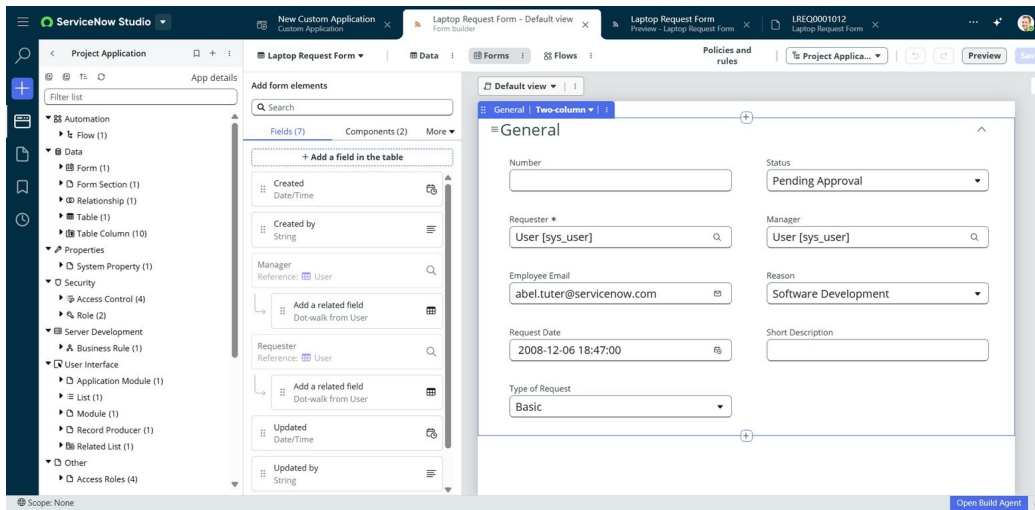


Figure 7.2 — ServiceNow Studio — Laptop Request Form fields (Form Builder view)

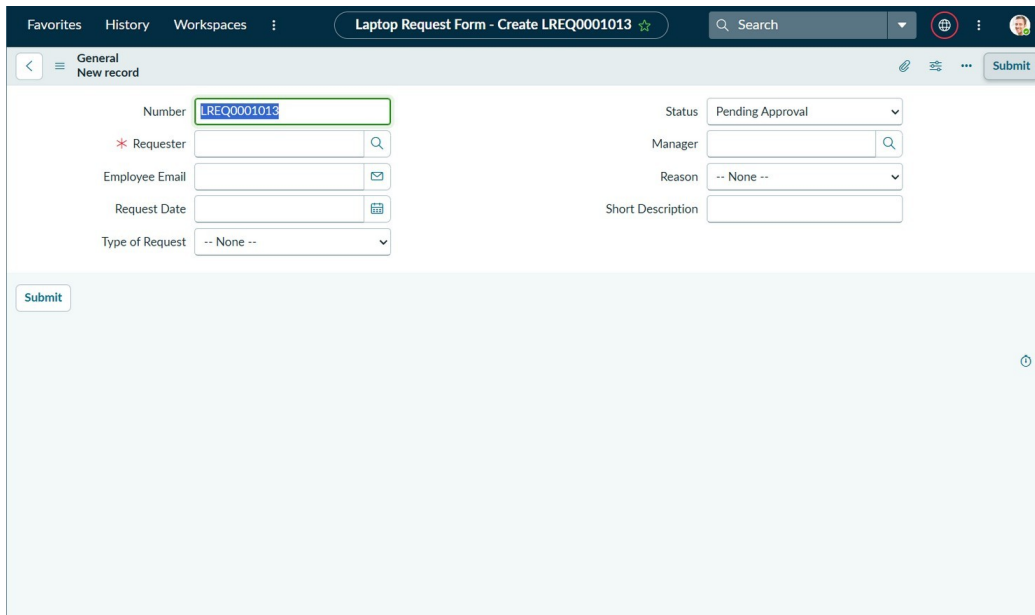


Figure 7.3 — Laptop Request Form — new record creation view

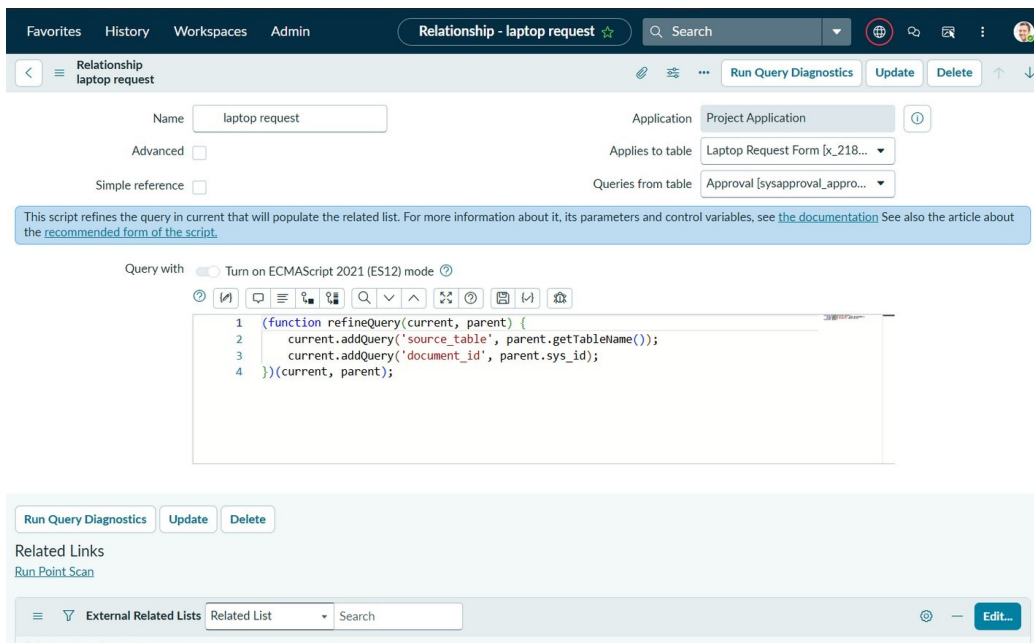


Figure 7.4 — Relationship record "laptop request" — script linking to the Approval table

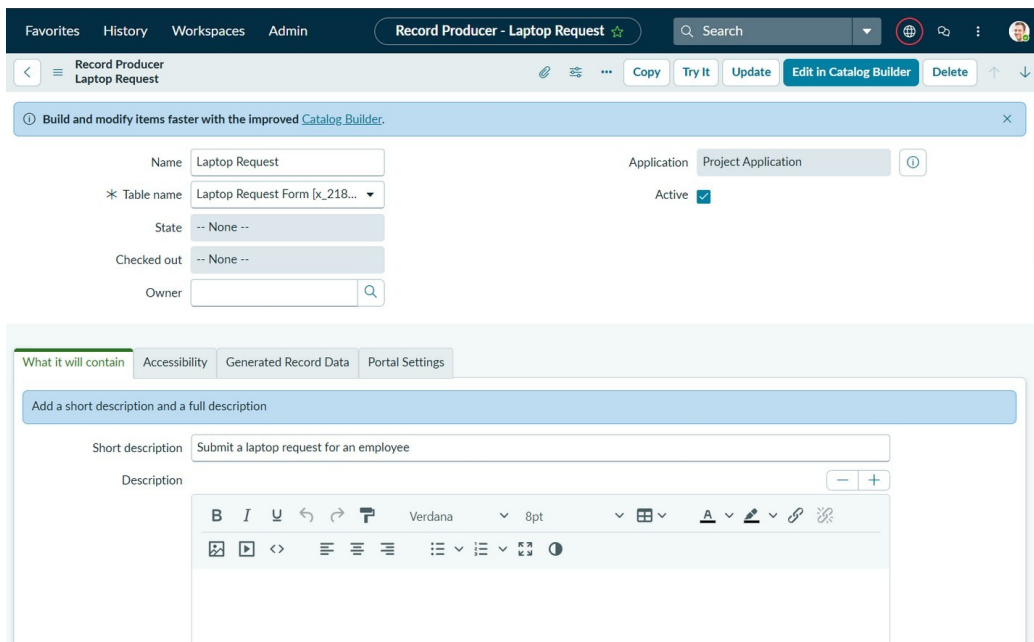


Figure 7.5 — Record Producer "Laptop Request" — main configuration

Variables (8)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

🔍

Order ▾

Search

🔗

—

Actions on selected rows... ▾

New

Catalog item = Laptop Request

☐	🔍	Question	Field	Type	Order ▴
		Requester	requester	Reference	100
		Employee Email	employee_email	Email	200
		Request Date	request_date	Date/Time	300
		Short Description	short_description	Single Line Text	400
		Manager	manager	Reference	500
		Type of Request	type_of_request	Select Box	600
		Status	status	Select Box	700
		Reason	reason	Select Box	800

⏪

⏩

1

to 8 of 8

⏪

⏩

🔗

Figure 7.6 — Record Producer — Variables list (8 variables mapped to table fields)

Employee Center is available to you
Join your peer organizations in creating a better employee experience with Employee Center, ServiceNow's new employee portal.

Learn More about Employee Center

servicenow

Knowledge Catalog Requests System Status Cart Tours System Administrator

Home Search laptop request

Sources
All Knowledge Bases Catalogs

FILTERS
Knowledge Bases Clear All
Category Apply
Author
Last modified
View Count
Catalogs Clear All
Category Apply

All results for "laptop request"

Laptop Request
Submit a **laptop request** for an employee

Install Software
Request for software installation service

What to do when you are locked out of your computer ?
Symptoms You can't log in to your laptop at the login screen Cause In the last 30 minutes you might have entered your password wrong at least 5 times * 2 possible resolutions : 1. Wait 30 Minutes for
Article: KB0005012 · Published: 9y ago

Dealing with Spyware and Viruses
· OIT advises that novice users request technical assistance when installing and using these programs

Figure 7.7 — "Laptop Request" catalog item visible in Service Portal search

B. Workflow & Automation

These screenshots show the Flow Designer approval workflow and the Business Rule that enforces the one-request-per-year policy.

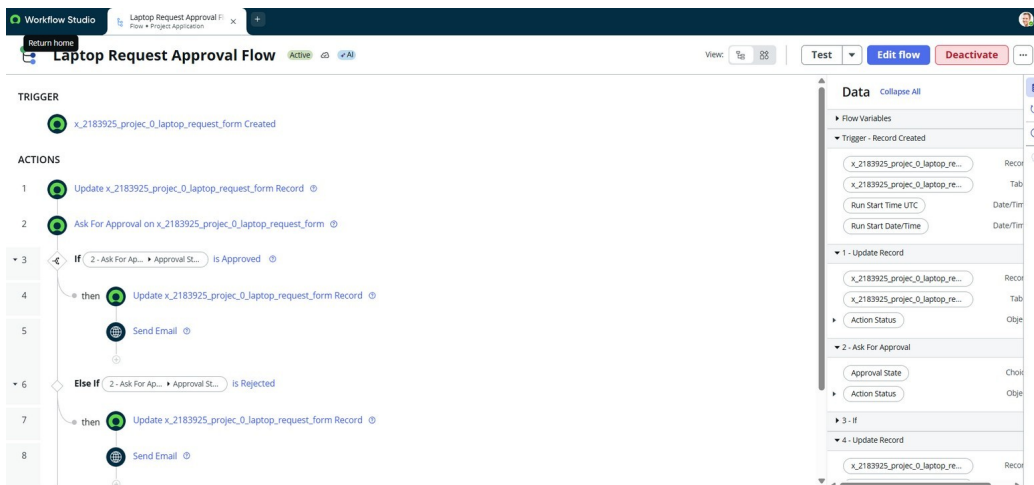


Figure 7.8 — Flow Designer — "Laptop Request Approval Flow", Active

Business Rule - Laptop request BR

Name: Laptop request BR

Table: Laptop Request Form [x_2183925_projec...]

Application: Project Application

Active: ☒

Advanced: ☒

When to run: before

Order: 100

Filter Conditions: Add Filter Condition Add OR Clause

Role conditions: [edit icon]

Insert: ☒

Update: ☐

Delete: ☐

Query: ☐

Update Delete

Related Links

Figure 7.9 — Business Rule "Laptop request BR" — When to run (Before, Insert)

Business Rule - Laptop request BR

Name: Laptop request BR

Table: Laptop Request Form [x_2183925_projec...]

Application: Project Application

Active: ☒

Advanced: ☒

When to run: before

Order: 100

Filter Conditions: Add Filter Condition Add OR Clause

Role conditions: [edit icon]

Script: Turn on ECMAScript 2021 (ES12) mode

```

1 (function preventDuplicateRequest(current, previous) {
2   var requester = current.getValue('requester');
3   var now = new GlideDateTime();
4   var currentYear = now.getYearOfC();
5   var startOfYear = currentYear + '-01-01 00:00:00';
6   var endOfYear = currentYear + '-12-31 23:59:59';
7
8   var gr = new GlideRecord('x_2183925_projec_0_laptop_request_form');
9   gr.addQuery('requester', requester);
10  gr.addQuery('sys_created_on', '>=' + startOfYear);
11  gr.addQuery('sys_created_on', '<=' + endOfYear);
12  gr.query();
13
14  if (gr.hasNext()) {
15    gs.addErrorMessage('You have already submitted a laptop request for this year.');
16    current.setAbortAction(true);
17  }
18  })(current, previous);
19

```

Update Delete

Figure 7.10 — Business Rule — Advanced script (duplicate-request check)

C. Security

These screenshots confirm the role-based access model: two custom roles, two groups with the correct role and membership, and four ACLs enforcing read/create/write permissions on the Laptop Request Form table.

Name	Description	Elevated privilege
x_2183925_projec_0_it_employee	IT Employee role - create and manage laptop requests	false
x_2183925_projec_0_it_manager	IT Manager role - view-only access to laptop requests	false

Figure 7.11 — Roles — it_manager and it_employee

Favorites History Workspaces Admin **Group - IT Employees** Search

< Group IT Employees Update Delete

Name Group email
 Manager Parent
 Description Who request for laptop

Update Delete

Roles (1) Group Members (10) Groups

Created Search Actions on selected rows... Edit...

Group = IT Employees

Created	Role	Granted by	Inherits
2026-07-29 01:59:38	x_2183925_projec_0_it_employee	(empty)	true

1 to 1 of 1

Figure 7.12 — Group "IT Employees" — Roles tab

Favorites History Workspaces Admin **Group - IT Employees** Search

< Group IT Employees Update Delete

Name Group email
 Manager Parent
 Description Who request for laptop

Update Delete

Roles (1) Group Members (10) Groups

User Search Actions on selected rows... New Edit...

Group = IT Employees

☐	Andrew Jackson
☐	Alicia Mountjoy
☐	Alyssa Biasotti
☐	Allie Pumphrey
☐	Alva Pennington
☐	Amos Linnan
☒	Andrew Och
☐	Allan Schwandt
☐	Amelia Caputo
☐	Allyson Gillisple

1 to 10 of 10

Figure 7.13 — Group "IT Employees" — Group Members (10 users)

Favorites History Workspaces Admin **Group - IT Managers** Search

< Group IT Managers Update Delete

Name Group email
 Manager Parent
 Description Group for IT managers who approve laptop requests

Update Delete

Roles (1) Group Members (8) Groups

Created Search Actions on selected rows... Edit...

Group = IT Managers

Created	Role	Granted by	Inherits
2026-07-29 01:56:40	x_2183925_projec_0_it_manager	(empty)	true

1 to 1 of 1

Figure 7.14 — Group "IT Managers" — Roles tab

Group - IT Managers

Name: IT Managers

Manager: [Search]

Group email: [Search]

Parent: [Search]

Description: Group for IT managers who approve laptop requests

Update Delete

Roles (1) Group Members (8) Groups

User Search

Group: IT Managers

User

- Adeia Cervantsz
- Alene Rabeck
- Abraham Lincoln
- Alejandro Mascall
- Abel Tuter
- Alejandra Prenatt
- Aileen Mottern**
- Alfonso Griglen

1 to 8 of 8

Figure 7.15 — Group "IT Managers" — Group Members (8 users)

Name	Decision Type	Operation	Type	Active	Updated by	Updated
x_2183925_projec_0_laptop_request_form	Allow If	create	record	true	admin	2026-07-29 00:16:56
x_2183925_projec_0_laptop_request_form	Allow If	read	record	true	admin	2026-07-29 00:16:56
x_2183925_projec_0_laptop_request_form	Allow If	read	record	true	admin	2026-07-29 00:16:56
x_2183925_projec_0_laptop_request_form	Allow If	write	record	true	admin	2026-07-29 00:16:56

Figure 7.16 — Access Control list — 4 ACLs on Laptop Request Form

Access Control - x_2183925_projec_0_laptop_request_form

Type: record

Application: Project Application

Operate: true

Active: ☒

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

Name: x_2183925_projec_0_laptop_request_form

Description: [Empty]

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

More info

Requires role

Role

x_2183925_projec_0_it_employee

Security Attribute Condition

Local or Existing: Local

Condition: (empty)

Figure 7.17 — ACL detail — create, role it_employee

Access Control - x_2183925_projec_0_laptop_request_form

Type: record
 Operation: read
 Admin overrides: ☒
 Decision Type: Allow if
 Protection policy: -- None --

Application: Project Application
 Active: ☒
 Advanced: ☐

Name: x_2183925_projec_0_laptop_request_form
 Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role

Role: x_2183925_projec_0_it_employee

Security Attribute Condition

Local or Existing: Local
 Condition: (empty)

Figure 7.18 — ACL detail — read, role it_employee

Access Control - x_2183925_projec_0_laptop_request_form

Type: record
 Operation: read
 Admin overrides: ☒
 Decision Type: Allow if
 Protection policy: -- None --

Application: Project Application
 Active: ☒
 Advanced: ☐

Name: x_2183925_projec_0_laptop_request_form
 Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role

Role: x_2183925_projec_0_it_manager

Security Attribute Condition

Local or Existing: Local
 Condition: (empty)

Data Condition

Figure 7.19 — ACL detail — read, role it_manager

Access Control - x_2183925_projec_0_laptop_request_form

Type: record
Operation: write
Application: Project Application
Active: ☒
Admin overrides: ☒
Decision Type: Allow if
Protection policy: -- None --
Name: x_2183925_projec_0_laptop_request_form
Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role: 1 to 1 of 1

Role
x_2183925_projec_0_it_employee

Security Attribute Condition

Local or Existing	Local
Condition	(empty)

Figure 7.20 — ACL detail — write, role it_employee

D. Functional Test Evidence

These screenshots are direct evidence from the live test cycle: a request that was approved, a request that was rejected, and the corresponding email notifications for each outcome.

Laptop Request Form - LREQ0001004

Number: LREQ0001004
 * Requester: Andrew Jackson
 Employee Email: abc@mail.com
 Request Date: 2026-07-30 21:36:41
 Type of Request: Basic
 Status: Approved
 Manager: Abel Tuter
 Reason: Personal Use
 Short Description: hi bye

Update Delete

State	Approver	Comments	Approval for	Created
Approved	Abel Tuter		(empty)	2026-07-29 21:37:28

Figure 7.21 — Request LREQ0001004 — Status Approved, Approvals related list

State	Approver	Comments	Approval for	Created
Approved	Abel Tuter		(empty)	2026-07-29 21:37:28
Rejected	Abel Tuter		(empty)	2026-07-29 02:20:22

Figure 7.22 — Approvals list — Approved and Rejected outcomes for Abel Tuter

Favorites History Workspaces Admin **Email - Laptop Request LREQ0001004 Approved** Search

Email Laptop Request LREQ0001004 Approved Update Delete

Type: UID:
 Target: Created:
 User: Deleted: ☐
 Notification type: Weight:
 Importance:

Subject:
 Recipients:
 Body:
 Content type:
 Headers:

Figure 7.23 — Email notification — Laptop Request LREQ0001004 Approved

Favorites History Workspaces Admin **Email - Laptop Request LREQ0001001 Rejected** Search

Email Laptop Request LREQ0001001 Rejected Update Delete

Type: UID:
 Target: Created:
 User: Deleted: ☐
 Notification type: Weight:
 Importance:

Subject:
 Recipients:
 Body:
 Content type:
 Headers:

Figure 7.24 — Email notification — Laptop Request LREQ0001001 Rejected

E. Deployment

These screenshots show the completed Update Set, confirming that every customization made across the project — table, fields, relationship, record producer, flow, business rule, roles, and ACLs — was captured for deployment.

Update Set - HR Onboarding Application							
Customer Updates (52)							
Update Set Logs							
Child Update Sets							
Install History							
Update set = HR Onboarding Application							
Created	Type	View	Target name	Updated by	Remote update set	Action	
2026-07-29 00:16:50	Variable		Reason	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:50	Variable		Requester	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:50	Variable		Employee Email	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:50	Variable		Manager	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:50	Variable		Type of Request	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Variable		Short Description	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Variable		Status	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Variable		Request Date	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Rejected	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Personal Use	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Advance	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Basic	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Other	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Software Development	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Pending Approval	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Question Choice		Approved	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Catalog Items Catalog			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Catalog Item Category			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:51	Record Producer		Laptop Request	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:58	Application Information			admin	(empty)	INSERT_OR_UPDATE	

Figure 7.25 — Update Set Customer Updates — records 1–20 of 52

Update Set - HR Onboarding Application							
Customer Updates (52)							
Update Set Logs							
Child Update Sets							
Install History							
Update set = HR Onboarding Application							
Created	Type	View	Target name	Updated by	Remote update set	Action	
2026-07-29 00:16:51	Choice list		x_2183925_projec_0_laptop_request_form.reason	admin	(empty)		
2026-07-29 00:16:52	Choice list		x_2183925_projec_0_laptop_request_form.status	admin	(empty)		
2026-07-29 00:16:52	Choice list		x_2183925_projec_0_laptop_request_form.type_of_request	admin	(empty)		
2026-07-29 00:16:52	Table		Laptop Request Form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:52	Dictionary		x_2183925_projec_0_laptop_request_form.employee_email	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:52	Dictionary		x_2183925_projec_0_laptop_request_form.manager	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:54	Dictionary		Laptop Request Form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:54	Dictionary		Laptop Request Form.Number	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:54	Dictionary		Laptop Request Form.Reason	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:55	Dictionary		Laptop Request Form.Requester	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:54	Dictionary		Laptop Request Form.Request Date	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:55	Dictionary		Laptop Request Form.Short Description	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Dictionary		Laptop Request Form.Status	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Dictionary		Laptop Request Form.Type of Request	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Flow		Laptop Request Approval Flow	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Number		LREQ	admin	(empty)	INSERT_OR_UPDATE	
2026-07-30 01:00:10	System Property		glide.db.instant_alter.enabled	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 22:17:46	Relationship		laptop request	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Business Rule		Laptop request BR	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Access Control		x_2183925_projec_0_laptop_request_form	admin	(empty)	INSERT_OR_UPDATE	

Figure 7.26 — Update Set Customer Updates — records 21–40 of 52

Customer Updates (52)							
Update Set Logs							
Child Update Sets							
Install History							
Update set = HR Onboarding Application							
Created	Type	View	Target name	Updated by	Remote update set	Action	
2026-07-29 00:16:56	Access Control		x_2183925_projec_0_laptop_request_form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Access Control		x_2183925_projec_0_laptop_request_form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Access Control		x_2183925_projec_0_laptop_request_form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Access Roles			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:56	Access Roles			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Access Roles			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Access Roles			admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Form Sections		Laptop Request Form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 22:20:05	Related Lists	Default view	Laptop Request Form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Form Layout		Laptop Request Form	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Role		x_2183925_projec_0_lt_manager	admin	(empty)	INSERT_OR_UPDATE	
2026-07-29 00:16:57	Role		x_2183925_projec_0_lt_employee	admin	(empty)	INSERT_OR_UPDATE	

Figure 7.27 — Update Set Customer Updates — records 41–52 of 52

8. Advantages & Disadvantages

As with any first version of a system, the application delivers clear, immediate benefits while also leaving a few areas that are explicitly scoped for a follow-up iteration rather than silently ignored.

8.1 Advantages

- Eliminates manual, email-based laptop requests with a standardized digital workflow.
- Automated manager approval removes follow-up emails and delays.
- Business Rule enforcement prevents duplicate requests without manual policing.
- Role-based ACLs ensure managers and employees see only what they should.
- Built entirely on the organization's existing ServiceNow instance — no new tool cost.
- Update Set packaging makes the solution portable to other instances.

8.2 Disadvantages

- Currently limited to laptop requests; not generalized to other asset types without further configuration.
- Duplicate-request and full ACL-enforcement test cases still require a live execution cycle before final sign-off.
- Email delivery depends on the instance's configured outbound mail account (not verified for real-world delivery in a Developer Instance).

9. Conclusion

This project successfully replaced a manual, email-driven laptop request process with a centralized, automated ServiceNow application. Employees can submit requests through a self-service catalog item, managers are automatically notified and can approve or reject with a single action, and the system enforces both eligibility (one request per year) and access-control policy without manual oversight. End-to-end functional testing confirmed the submission-to-approval and submission-to-rejection paths both work correctly, with email notifications delivered at each stage.

Across the full build — data model, workflow automation, security, and testing — the project demonstrates that a meaningful piece of enterprise IT process automation can be delivered natively on ServiceNow without any custom code outside of a single business rule script and a relationship query, using the platform's own catalog, flow, and security building blocks.

10. Future Scope

The following enhancements are recommended as the next iteration of this application:

- Extend the catalog to cover other IT assets (monitors, peripherals, software licenses).
- Add a manager dashboard/report showing team-wide request history and turnaround time.
- Integrate with IT Asset Management (ITAM) to automatically reserve inventory upon approval.
- Add SLA timers so overdue approvals automatically escalate to a secondary approver.
- Complete the duplicate-request and ACL-enforcement live test cycle for full sign-off.

11. Appendix

Source Code: N/A — this project is a no-code/low-code ServiceNow configuration (Table, Flow, Business Rule, ACLs) rather than a traditional codebase. The complete configuration is packaged in the "HR Onboarding Application" Update Set (52 customizations, exportable to XML).

Dataset Link: N/A — no external dataset is used; all data is generated within the ServiceNow instance through user-submitted requests.

GitHub & Project Demo Link: _____ (add your repository / demo link here)